Lab4 (1.1).py - C:/Users/Vaishnavi PM/AppData/Local/Programs/Python/Python314/Lab4 (1.1).py (3.14.2)

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```python
import numpy as np
a1=np.array([1,2,3,4,5,6,7,8,9])
print(a1)
a=a1.reshape(3,3)
print(a)
print("accessing")
print(a[2][1])
print(a[0:2])

print("all row 2nd col")
print(a[:,2])

print("modify")
a[2][1]=60
print(a)

print("\n")
sum=np.sum(a)
print(sum)

print("sum of col")
sumc= np.sum(a,axis=0)
print(sumc)

print("sum of rows")
sumr= np.sum(a,axis=1)
print(sumr)

print("addition and multiplication to each ele")
a=np.array([1,2,3])
print(a+2)
print(a*3)

print("operations of two matrics")
a = np.array([10, 20, 30])
b = np.array([1, 2, 3])
print(a + b)
print(a - b)
print(a * b)

print("special arrays")
zeros=np.zeros((2,3))
ones=np.ones((2,3))
eye=np.eye(3)
arange=np.arange(0,10,2)
linspace=np.linspace(0,1,3)
```

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```
sumc= np.sum(a,axis=0)
print(sumc)

print("sum of rows")
sumr= np.sum(a,axis=1)
print(sumr)

print("addition and multiplication to each ele")
a=np.array([1,2,3])
print(a+2)
print(a*3)

print("operations of two matrics")
a = np.array([10, 20, 30])
b = np.array([1, 2, 3])
print(a + b)
print(a - b)
print(a * b)

print("special arrays")
zeros=np.zeros((2,3))
ones=np.ones((2,3))
eye=np.eye(3)
arange=np.arange(0,10,2)
linspace=np.linspace(0,1,3)
print(zeros)
print(ones)
print(eye)
print(arange)
print(linspace)

print("build in functions")
a=np.array([10, 20, 30, 40])
print(np.sum(a))
print(np.mean(a))
print(np.median(a))
print(np.max(a))
print(np.min(a))
```

IDLE Shell 3.14.2

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```
>>>
================================= RESTART: C:/Users/Vaishnavi PM/AppData/Local/Programs/Python/Python314/Lab4 (1.1).py =================================
[1 2 3 4 5 6 7 8 9]
[[1 2 3]
 [4 5 6]
 [7 8 9]]
accessing
8
[[1 2 3]
 [4 5 6]]
all row 2nd col
[3 6 9]
modify
[[ 1  2  3]
 [ 4  5  6]
 [ 7 60  9]]


97
sum of col
[12 67 18]
sum of rows
[ 6 15 76]
addition and multiplication to each ele
[3 4 5]
[3 6 9]
operations of two matrics
[11 22 33]
[ 9 18 27]
[10 40 90]
special arrays
[[0. 0. 0.]
 [0. 0. 0.]]
[[1. 1. 1.]
 [1. 1. 1.]]
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
[0 2 4 6 8]
[0.  0.5 1. ]
build in functions
100
25.0
25.0
40
10
>>>
```

Ln: 374 Col: 0

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> import array
>>> arr=array.array('i', [10,100,30])
>>> print(arr)
array('i', [10, 100, 30])
>>> arr.append (50)
>>> print(arr)
array('i', [10, 100, 30, 50])
>>> #arr.sort() # List sort() method wont work #print(arr)
>>> # use array method
>>> arr=array.array('i', sorted(arr))
>>> print(arr)
array('i', [10, 30, 50, 100])
>>>
```

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> import numpy as np
>>> arr1=np.array([1,2,3,4,5,6])
>>> print(arr1)
[1 2 3 4 5 6]
>>> import numpy as np
>>> arr1=np.array([1,2,3,4])
>>> print(arr1)
...
[1 2 3 4]
>>> rr2d=np.array([ [1,2], [3,4]])#2D
...
>>> print(rr2d)
[[1 2]
 [3 4]]
>>>
```